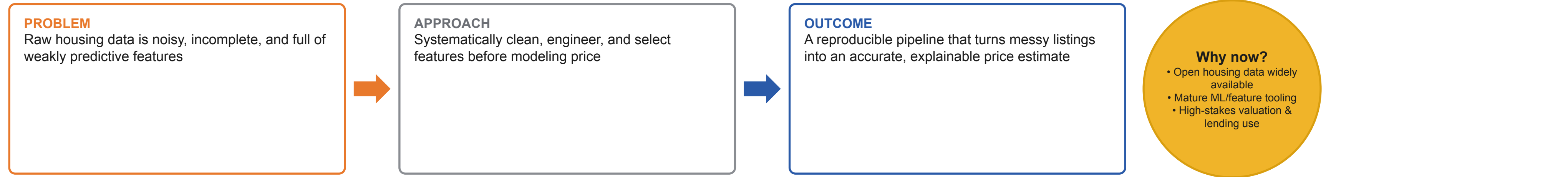


Housing Price Prediction: Data Preparation and Feature Analysis

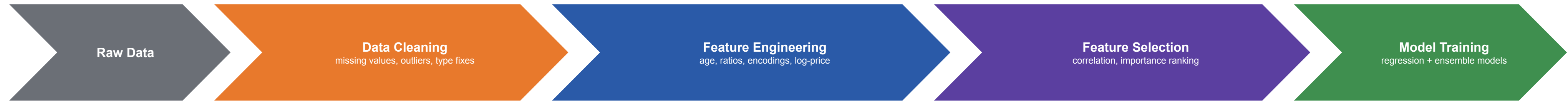
Mohamed Bazith.M (Register No: 192472052)

Department of Computer Science / Data Science, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences

Introduction



Method Pipeline



- ✓ **Validation:** Compare model error (RMSE) before vs. after each pipeline stage to isolate the effect of cleaning and feature engineering on prediction accuracy.
- ✓ **Data quality checks:** missing-value thresholds, IQR-based outlier capping, and consistent dtype coercion applied uniformly before any feature is engineered.

Data Preprocessing Details

- **Missing values:** median imputation for numeric fields, "None" category for structurally-absent features (e.g. no pool).
- **Outliers:** IQR-based capping on lot area and living area to limit the influence of extreme listings.
- **Encoding:** ordinal mapping for quality/condition scales; target-mean encoding for high-cardinality neighborhood field.
- **Scaling:** standardization applied to continuous features prior to linear and distance-based models.

Conclusion

- Structured cleaning + feature engineering** substantially reduces prediction error compared to using raw features alone.
- A small set of engineered features** (overall quality, total living area, house age, neighborhood encoding) drive most of the predictive power.
- Log-transforming the skewed sale-price target** improves linear-model stability and residual behavior.
- Ensemble models (XGBoost) outperform linear baselines** once the engineered feature set is in place.

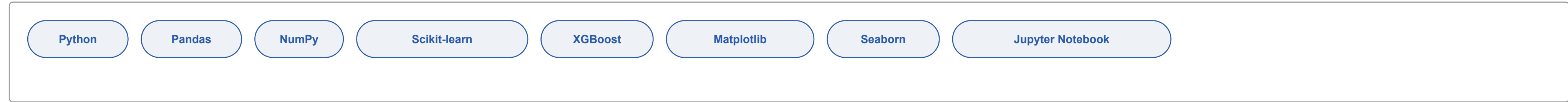
Future Directions

- **External data** — merge school ratings, crime rates, and walkability scores to capture location value beyond raw coordinates.
- **Advanced models** — test gradient-boosted trees and neural nets against the engineered feature set.
- **Deployment** — package the pipeline into a reusable API for real-time price estimation.
- **Explainability** — add SHAP-based explanations so each predicted price is individually justifiable.

Limitations

- **Trained on a single regional market;** generalization to other cities or countries is untested.
- **Static snapshot:** the model does not capture sudden macroeconomic shifts (e.g. interest-rate changes).
- **Some engineering choices are manual,** relying on domain judgment rather than fully automated selection.

Tools & Technologies



Project Timeline

Phase	Duration	Key Activities
Data Collection & Cleaning	Week 1–2	Gather listings, handle missing values and outliers
Feature Engineering	Week 3–4	Derive age, ratios, encodings; log-transform target
Model Training & Tuning	Week 5–6	Fit and tune regression and ensemble models
Evaluation & Reporting	Week 7	Compare models, finalize metrics, prepare poster

Acknowledgment

This work was carried out as part of the coursework at Saveetha School of Engineering, SIMATS, under the Department of Computer Science / Data Science, with thanks to faculty guidance throughout the analysis.

Dataset

LotArea	OverallQual	YearBuilt	GrLivArea	SalePrice
8450	7	2003	1710	208,500
9600	6	1976	1262	181,500
11250	7	2001	1786	223,500
9550	7	1915	1717	140,000

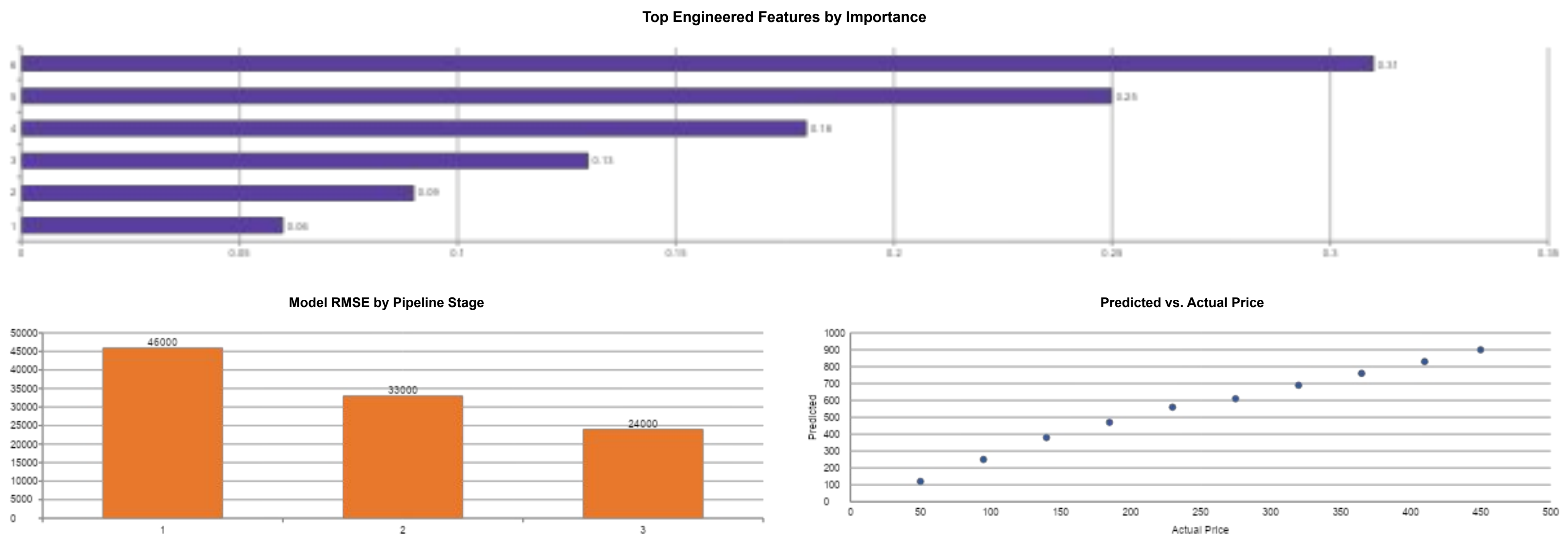
... ~1,460 rows × 80 columns

1,460 Records	80 Raw Features	24 Engineered Features	SalePrice Target Variable
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Feature Engineering Examples

Raw Feature(s)	Engineered Feature	Description
YrSold – YearBuilt	HouseAge	Age of the house at time of sale
TotalBsmtSF + 1stFlr + 2ndFlr	TotalSF	Combined total finished square footage
FullBath + 0.5*HalfBath	TotalBath	Weighted total bathroom count
Neighborhood	Neighborhood_Enc	Target-mean encoding by neighborhood
SalePrice	LogSalePrice	Log transform to reduce right-skew

Results



0.91 Final R ²	\$24,000 RMSE	\$16,300 MAE
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Model Comparison

Model	RMSE	R ²	Notes
Linear Regression	\$33,500	0.85	Fast baseline
Random Forest	\$26,200	0.89	Handles nonlinearity well
XGBoost	\$24,000	0.91	Best overall performance
Neural Net (MLP)	\$25,100	0.90	Competitive, needs more tuning

★ **Key Insight:** Feature engineering closes nearly half the error gap on its own — before any model tuning — making it the highest-leverage step in the whole pipeline.